DPR Requirements [VIL-Instance]

ELFOUR-1730 - Analysis of requirements of 3BU_17300_1257_DTAPA chapter 3.1.4.3.2. VIL Availability OPEN

- Analysis of requirements of 3BU_17300_1257_DTAPA chapter 3.1.4.3.2. VIL Availability
- Furthermore of this requirements must be allocated down to VIL SW Components.

Input from: https://dwf.at.gts/elektra/details/3BU_17300_1257_DTAPA.01_01

Requirement ID	Description	Synopsis	Availability Class /Value	Allocation Implication
EL4-DIL-SSS-REQ-00587	Availability - [VIL-Instance] - Overall	The [VIL-Instance] shall have a class 2 overall availability of at least 99.995 %.	Class 2 (0.99995)	General structural requirement for both VSTW Core and VSTW MUX.
EL4-DIL-SSS-REQ-00569	**Availability distribution of [VIL-Instance]**	The distribution of the availability of a [VIL-Instance] must correspond to class 2. This means in particular an availability of 99.9979 % (MTTR = 0.5 h) 99.999 % (MTTR = 10 min) 99.9996 % (MTTR = 2 min) Comment: Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.	Class 2	Depends on redundancy architecture and support /diagnostic systems.
EL4-DIL-SSS-REQ-00570	Availability - [VIL-Instance] - **Outages** with no operational Impact	Outages of a [VIL-Instance] with no operational impact shall have an availability of at least 0,997154. Comment: Outages of a [VIL-Instance] with no operational impact correspond to defect class FK4.	Class 4	
EL4-DIL-SSS-REQ-00571	Auto-correction within 10 minutes	delted and merged with EL4-DIL-SSS-REQ-00569	Class 2 (99.999 %)	Requires rapid detection and recovery mechanisms in Core logic.
EL4-DIL-SSS-REQ-00572	Auto-correction within 2 minutes	delted and merged with EL4-DIL-SSS-REQ-00569	Class 2 (99.9996 %)	Highest availability requirement, critical for redundant operation (Core and MUX).
EL4-DIL-SSS-REQ-00573	Minor SW Updates (Patches)		No operational restrictions	Requires Hot-Hot redundancy handover capabilities (MUX).
EL4-DIL-SSS-REQ-00818	Major SW Updates (Upgrades)		Carried out during operational breaks	Requires seamless takeover by redundancy during updates (Core /MUX).
EL4-DIL-SSS-REQ-00574	Operational Downtime Limit		Startup time + re-integration time	Requires efficient software management mechanisms (ADM/MUX).

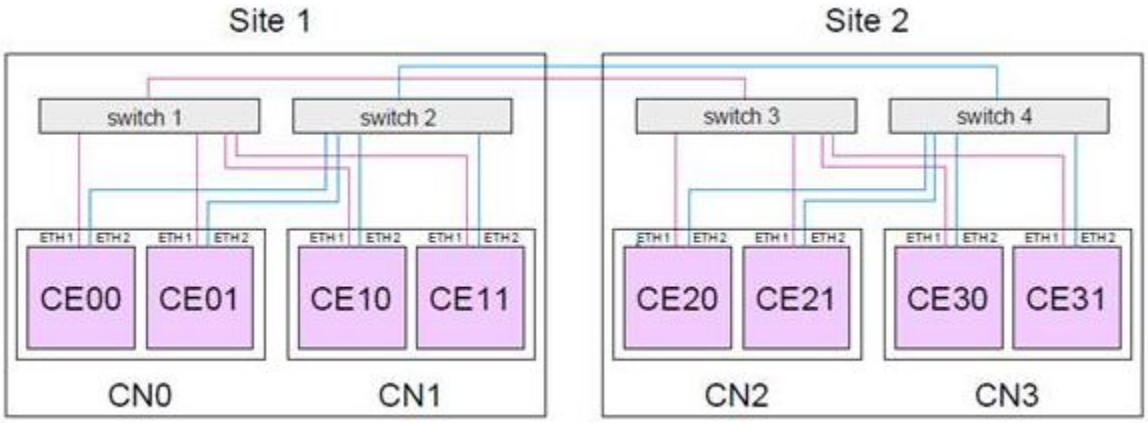
VSTW Core System

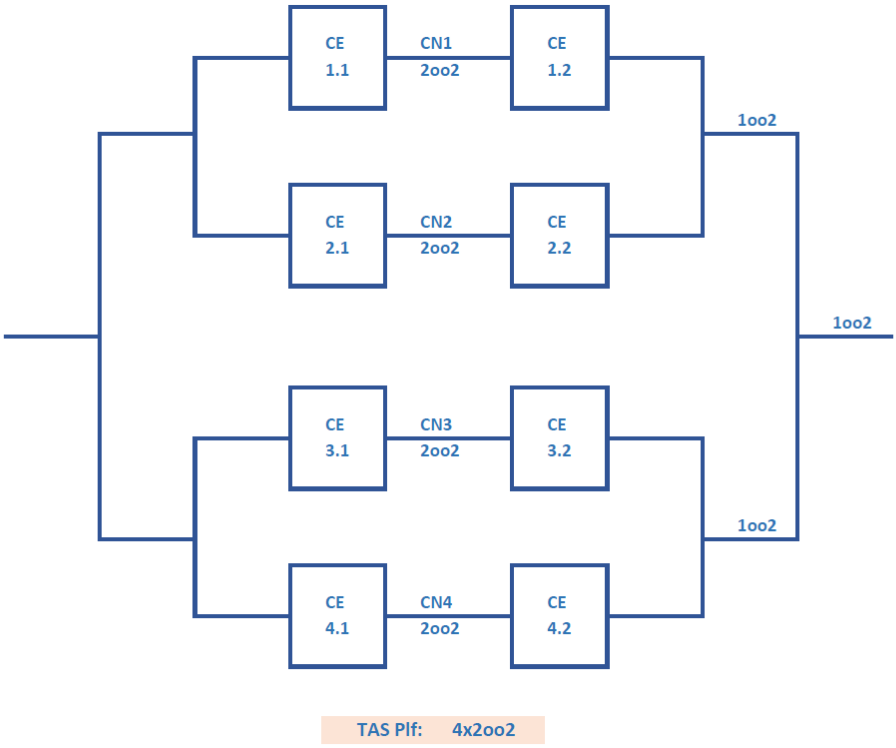
Requirement(s)	Allocation Justification (VSTW Core)
EL4-DIL-SSS-REQ-00587, EL4-DIL-SSS-REQ-00569	The high overall and seamless auto-correction availability values rely heavily on the Hot-Hot redundancy architecture. If one Hot-Hot redundant system fails, operation continues without impact, and the failed system can be reintegrated during runtime. This seamless operation is key to achieving Class 2 availability targets for quick recovery.
EL4-DIL-SSS-REQ-00570 (Errors without operational impact)	This requires the core logic to be robust enough so internal errors do not compromise the system's primary function. The two diversitary software channels within each 2oo2 Computing Node (CN) ensure hardware faults are safely detected.

EL4-DIL-SSS-REQ-00569 **Availability distribution of [VIL-Instance]**	The definition of "Automatic correction" involves entering a safe state, performing a self-test, and automatic reintegration of the disturbed component. This process is inherent to the safe computing unit (CE/CN) defined within the VSTW Core architecture.
EL4-DIL-SSS-REQ-00818 (Major SW Upgrades)	The Core system must ensure redundancy takes over operation (02_259) to facilitate the update during an operational break. (EL4-DIL-SSS-REQ-00818)

VSTW MUX System

Requirement(s)	Allocation Justification (VSTW Core)
EL4-DIL-SSS-REQ-00587 (Overall Availability)	As the crucial component connecting the Core logic to the OCs/Field Element Stations (FES), the MUX must also meet the high overall availability standard. Since MUX redundancies run Hot-Hot, failure of one redundancy does not lead to operational restriction.
EL4-DIL-SSS-REQ-00573 (Minor SW Patches) & EL4-DIL-SSS-REQ-00574 (Downtime Limit)	Minor SW updates (Patches) are carried out *without operational restrictions*. This is done by the MUX's Hot-Hot redundancy, allowing one redundancy partner to be taken out of service, patched, and seamlessly reintegrated. This minimal downtime relies on the MUX managing the connection handover.
EL4-DIL-SSS-REQ-00569 **Availability distribution of [VIL-Instance]**	Met by the immediate, seamless failover capability provided by the Hot-Hot high-availability architecture:





VSTW Core

- The VSTW Core system uses a **2x2oo2 = 1oo2 or 4x2oo2 = 1oo4 Redundancy Concept**, running applications synchronously in a Hot-Hot configuration.
- If a component (CN/CE) fails, the remaining redundancies *must* continue operation without restriction. The logic for achieving self-test, safe state entry, and automatic recommissioning is a core function of the safe computation platform (CN/CE), which directly enables the VSTW application (VSTW Core) to meet this recovery deadline

VSTW MUX

- The MUX's functions are operated on the same hardware instances as the VSTW Core. If the underlying hardware fails, both Core and MUX instances on that hardware fail together.

SSS_DIL Requirements allocated to by SSS_VIL

The high overall availability requirement for the VIL (**EL4-DIL-SSS-REQ-00587**) is achieved through a hot-hot redundancy architecture. This is supported by several architectural and component-specific requirements in SSS_VIL.

The ability of the VIL to quick self-correct following a fault is crucial for achieving high availability values.

SSS_DIL Availability Requirement **3BU 17300 1257 DTAPA**	**SSS_VIL Redundancy & Architecture Requirements** **3BU 1500 6955 DTAPA**	**Info**	**Origin for SSS VIL** **SSS DIL CU** **3BU 17300 1255 DTAPA**
Requirements Ensuring High Overall Availability			

EL4-DIL-SSS-REQ-00587 (Overall Availability)	**EL4-VIL-SSS-REQ-00218** VIL Hot-Hot Redundancy (Basic)	VIL shall support up to two VIL redundant nodes running in **hot-hot redundancy** in the basic configuration	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy
	EL4-VIL-SSS-REQ-00219 VIL Hot-Hot Redundancy (Georedundant)	In georedundant configuration, VIL shall support up to **four VIL redundant nodes** running in hot-hot redundancy	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00457] DIL Variants - Geo-Redundancy - Different Servers ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00458] DIL Variants - Geo-Redundancy - Seamless Operation ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00456] DIL Variants - Geo-Redundancy - Different DC Locations
	EL4-VIL-SSS-REQ-00181 ELEKTRA Core - Fault Tolerance Architecture (Basic)	**ELEKTRA Core** (VIL central control) must provide hot-hot redundancy using the TAS Platform NMR service in a **2x2oo2 configuration**, ensuring robustness against any single Hardware (HW) failure affecting one Computing Node (CN)	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00443] VIL Redundancy Concept - Availability
	EL4-VIL-SSS-REQ-00227 EC-MUX Fault Tolerance Architecture (Basic)	**EC-MUX** shall provide hot-hot redundancy using the J4S fault tolerance service in a **2x1oo1 configuration**, robust against any HW failures affecting one redundant node	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00443] VIL Redundancy Concept - Availability
	EL4-VIL-SSS-REQ-00222 VIL Split-Brain Prevention	It shall be ensured that only one subgroup of synchronized nodes remains active when VIL nodes are disconnected, preventing the Split-Brain problem (critical for availability),	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy
Requirements for Automatic Fault Correction + Manual Correction			
EL4-DIL-SSS-REQ-00569 **Availability distribution of [VIL-Instance]**	**EL4-VIL-SSS-REQ-00221** VIL Automatic Reintegration	When a VIL redundant node starts while the VIL instance is already operational, it shall be **automatically integrated** (Recovery) into the VIL instance without service interruption. This directly supports the strict time limits (2 minutes / 10 minutes) for automatic correction specified.	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00453] DIL Variants - SW Updates
	EL4-VIL-SSS-REQ-01155 VIL Application Restart	In the event of a Software (SW) failure of one of the VIL SW applications, the VIL must automatically **stop and restart the affected application** on all redundant nodes. This fast recovery is necessary to meet auto-correction time targets.	no origin
	EL4-VIL-SSS-REQ-00285 VIL Diagnosis Processing - Redundancy	The VIL's diagnostic function must run fully synchronized on all redundant nodes, ensuring that fault detection (a prerequisite for auto-correction) remains available whenever at least one VIL redundant node is operational.	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00476] Maintenance and Diagnosis - Interface SDI-EIL
Requirements for Maintenance, Updates, and Downtime			

EL4-DIL-SSS-REQ-00573 (Minor SW updates without operational restrictions)	**EL4-VIL-SSS-REQ-00218** VIL Hot-Hot Redundancy (Basic)	The hot-hot redundancy ensures that updates can be applied to one node while the redundant partner continues operation seamless.	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy
	EL4-VIL-SSS-REQ-00221 VIL Automatic Reintegration	Necessary to seamlessly re-integrate a patched redundant node back into the operational VIL instance after updating.	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00439] VIL Redundancy Concept - Failure Detection ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00453] DIL Variants - SW Updates
EL4-DIL-SSS-REQ-00818 (Major SW Updates during operational breaks)	**EL4-VIL-SSS-REQ-01156** VIL Maintenance Shutdown	The VIL must provide a way to trigger a **controlled maintenance shutdown** of the complete VIL instance, including the graceful shutdown of all SW applications and subsequent OS shutdown on all Virtual Machines (VMs). This is required for major updates carried out during operational breaks.	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-01294] Activation of Software Updates -
EL4-DIL-SSS-REQ-00574 (Operational downtime limited to startup /reintegration time)	**EL4-VIL-SSS-REQ-00220** VIL Synchronized Startup of VIL Redundant Nodes	The VIL redundant nodes must be able to **start operation in a synchronized way**. This minimizes downtime during a full system reboot, linking directly to the timing constraints.	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00451] DIL Variants - Hot-hot Redundancy
Availability of Diagnosis Functionality			
EL4-DIL-SSS-REQ-00329 /00330 (DIL diagnosis of open/closed network failures)	**EL4-VIL-SSS-REQ-00244** RaSTA Communication - Connection Status	For each external RaSTA connection (used by OMC, RBC, ILS, etc.), VIL shall send diagnosis data concerning the connection status via the Datalogger Interface to the external diagnosis collector	Origin: SSS RAMS 3BU 17300 1257 DTAPA
	EL4-VIL-SSS-REQ-00245 RaSTA Communication - Transport Channel Status	For each external RaSTA connection, VIL must send diagnosis data detailing the status of each associated RaSTA transport channel	Origin: SSS RAMS 3BU 17300 1257 DTAPA
	EL4-VIL-SSS-REQ-00288 Datalogger Interface Redundancy	The datalogger interface shall have one transport channel per VIL redundancy node. This ensures availability of the diagnosis function even if only one VIL redundancy node remains operationa	SSS DIL CU ==> [EL4 SSS] [EL4-DIL-SSS-REQ-00450] DIL Variants - Basic Virtualisation

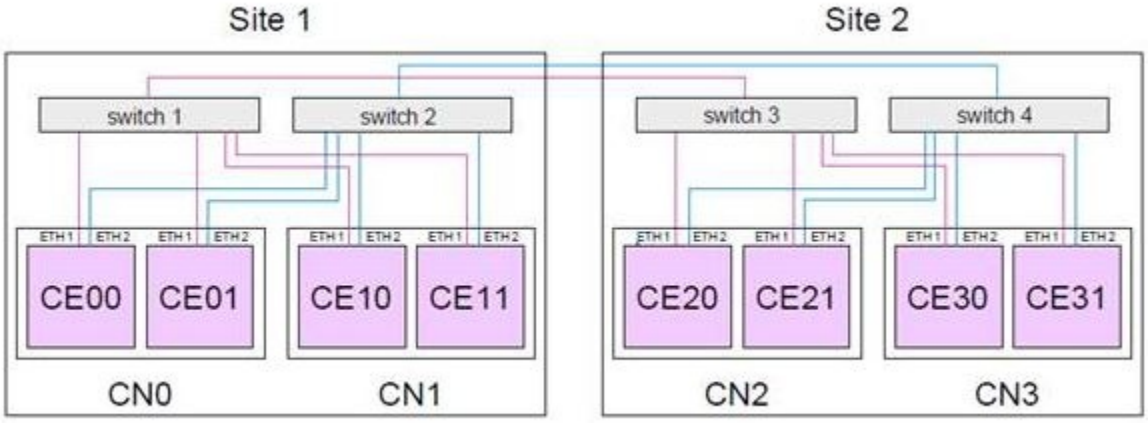
Automatic recovery

See also: 2025-11-11 VIL Performance Test - Session with RM and ARCH about corresponding SSS requirements

See: System Requirements Specification - Digital Interlocking OBB (DIL) - Central Unit Part 1 - 3BU 17300 1255 DTAPA Ed. 01

EL4-DIL-SSS-REQ-00991	**VIL Redundancy Concept - Recovery**	In case of a [failure] of a [VIL Node] each [VIL Node] shall carry out any necessary recovery of the affected [VIL Node]s **automatically.**

preliminary RAM calculations: Prelim_RAM_Estimation_DSTW_Ed01p11_20260116.xlsx show the following (see also VIL/VIL Instance)



CNs (CN0 und CN1) are **1oo2-redundant:**

preliminary RAM calculations give the following results:

Logical equations	FiT	MTBF [y]	Availability [%]	MTTR
CN1+CN2_1oo2_hot + Network - MTTR = 2min	4.449	25 658	99.999 999 99	2min
CN1+CN2_1oo2_hot + Network - MTTR = 5min	4.451	25 645	99.999 999 96	5min
CN1+CN2_1oo2_hot + Network - MTTR = 10min	4.455	25 623	99.999 999 92	10min
CN1+CN2_1oo2_hot + Network - MTTR = 1h	4.493	25 410	99.999 999 55	1h
CN1+CN2_1oo2_hot + Network - MTTR = 4h	4.627	24 671	99.999 998 14	4h

From RAM perspective it is reasonable to demand a "recovery time" for a VIL-Node of <10 minutes.